

A Gujarat-Sourced Sodium-Ion Anode

Presented By: InsightX



Problem Statement

Gujarat has abundant **agricultural biomass** and **sodium resources**, yet its sodium-ion battery ecosystem depends on **imported hard-carbon anodes**. The challenge is to convert **local agro-waste** into **battery-grade hard carbon** at **competitive cost** and **performance**.

Target Users

Primary: Sodium-ion cell & battery manufacturers

Secondary: Energy-storage, EV, and battery R&D companies

Biomass to Hard-Carbon

1. Raw Biomass Sourcing

(Cotton / Caster Stalks)
Machinery : Shredders
& Grinders

2. Demineralization & Ash Removal

Machinery: Glass-
lined Stirred Batch
Chemical Reactors

3. Pre-Oxidation & Stabilization

Machinery: Industrial
Air-Ventilated Rotary
Dryers

4. Pyrolysis & N/S Doping

Machinery: Calcination
Rotary Kilns &
Petrochemical Gas
Inlets

5. Post-Treatment & Milling

Machinery: Fluid Bed
Jet Mills & Vacuum
Shelf Dryers

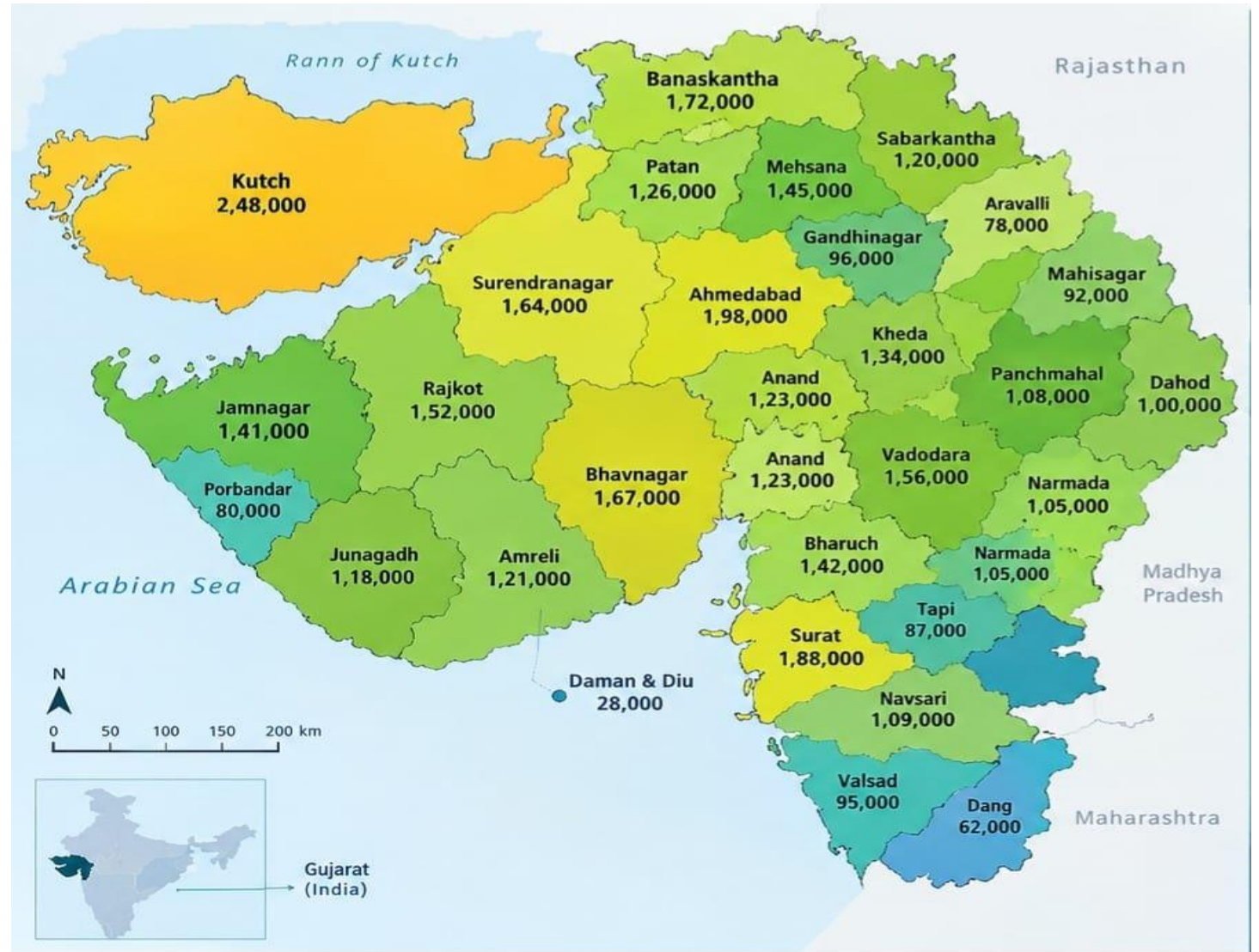
6. Hard Carbon

Anode Material
(Sodium-Ion Battery
Grade)

Biomass Data

~51.4 Million Tonnes /Year
Estimated gross agricultural
residue generated in Gujarat

- Feedstock Selection based on:
Availability + Ash content +
Carbon yield + Collection
feasibility + Transport distance

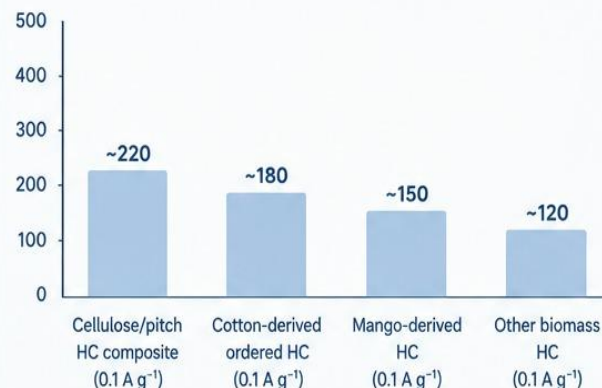


Effect of Heteroatom Doping on Hard Carbon Performance

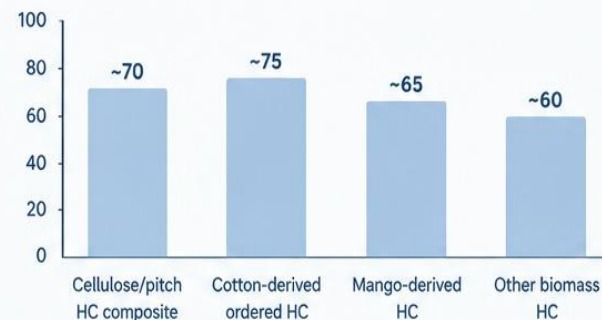
Doping (N / S) enhances sodium storage capacity, rate performance and initial Coulombic efficiency (ICE).

Before Doping (Undoped Hard Carbon)

Specific Capacity (mAh g⁻¹)

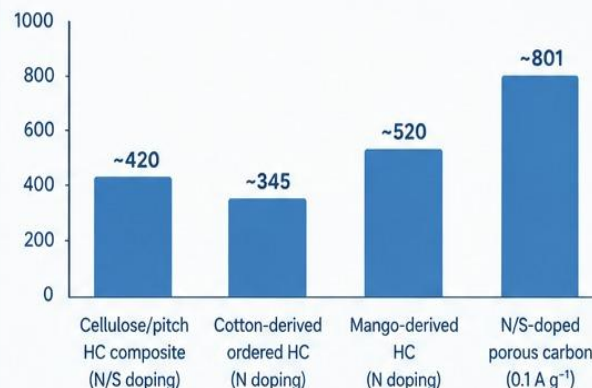


Initial Coulombic Efficiency (ICE) (%)

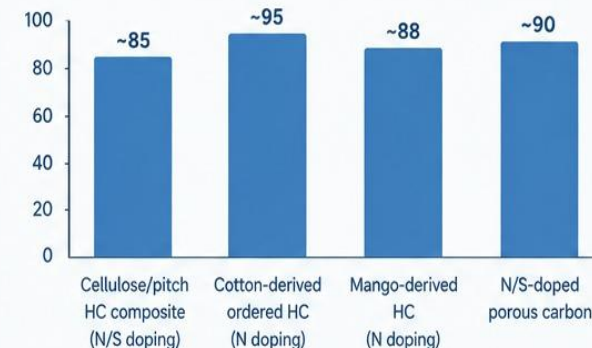


After Doping (N / S Doped Hard Carbon)

Specific Capacity (mAh g⁻¹)



Initial Coulombic Efficiency (ICE) (%)



Key Improvements with Doping



Higher Capacity

Doping introduces active sites and improves electronic conductivity.



Better ICE

Reduces irreversible reactions and stabilizes the SEI layer.



Faster Kinetics

Enhances Na⁺ diffusion and charge transport.



Improved Stability

Strengthens structure and cycle life.



Note

Values are from reported results in the reference paper and may vary based on biomass source, dopant type and synthesis conditions.

Source: ACS Nano 2024, 18, 16468–16488.
(Figures 7a, 7b and related discussion)

- Hard Carbon before processing : Limited active sites and unoptimized surface chemistry/interlayer structure can restrict Na⁺ storage and contribute to low ICE and poor rate performance.
- Hard-carbon performance can be enhanced through heteroatom doping, interlayer-spacing control, pore-structure optimization and carbonisation-temperature tuning.

Table 1. Summary of Classification and Properties of Biomass-Derived Hard Carbon Anodes for SIBs

Classification	Precursor	Preparation condition	Capacity (mAh/g)	ICE%	Cycle number/capacity
Animal and plant organs	Eucalyptus ⁴⁵	1400 °C/Ar/2h	239	89.1	200/93
	Switchgrass ¹⁵⁴	1000 °C/Ar/2h	225	42	400/96
		2050 °C/Ar/2h	210	64	800/~100
	Pomelo peel ⁶⁶	1200 °C/N ₂ /2h	430.5	68	200/97.5
	Date palm ¹⁴⁰	1200 °C/N ₂ /2h	300	88.4	10/78.3
	Moss ³⁴	1100 °C/Ar/and 300 °C activation 3h	298	57.5	210/85.6
	Pine pollen ¹⁵⁵	900 °C/Ar/1h	370	59.8	200/98
	Lotus root ¹⁵⁶	1400 °C/Ar	350	70	450/94
	Lotus seed ⁴⁷	1200 °C/Ar/1h	328.8	50.4	200/90
	Cucumber stem ¹⁵⁷	1000 °C/N ₂ /2h	337.9	64.9	500/82.1
	Eggshell membrane ¹⁵⁸	1300 °C/Ar/2h	310	89	250/99
	Lotus leaves ⁵⁸	hydrothermal acid 800 °C/Ar/2h	250	66	100/250
	Garlic ¹⁵⁹	1300 °C/N ₂ /6h	260	50.7	1000/105
	Cotton ³⁹	1300 °C/Ar/2h	315	83	100/305
	Carrageenan ¹⁴²	800 °C/Ar/1h and 600 °C/CO ₂	283	68.9	400/~100
Biological extracts	Wood fiber ¹⁶⁰	1000 °C/Ar/2h	240	72	200/~100
	Cellulose ¹⁴¹	1600 °C/Ar/2h	310	84	500/~100
	Glucose ¹⁴³	sulfuric acid, nitric acid and sodium carbonate	323.9	82.9	500/72.4
	Bacterial cellulose ¹⁶¹	800 °C/N ₂ /2h	223	76.1	1500/210
Biological waste	Peanut skin ¹⁴⁵	Hydrothermal pretreatment, 800 °C/Ar/1h	174–431	29–34	200/86
	Waste tea ¹⁶²	1400 °C/Ar/2h	282.4	69	200/83
	Sawdust ¹⁶³	900 °C/Ar/1h	275	56	500/69.8
	Lignin ¹⁴⁶	1600 °C/Inert gas under carbonization	303	81.2	
	Lignin/bitumen ⁵¹	1400 °C/Ar/2h	254	82	150/89
	Corn stalk ⁶⁵	Activation, 800 °C/Ar/1h	439	21.7	50/90
	Bagasse ¹⁶⁴	1000 °C/N ₂ /2h	242.1	73.1	800/91.5
	Reed straw ¹⁴⁴	Hydrothermal pretreatment, 1300 °C/N ₂ /3h	372	77.03	200/84

Gujarat Biomass → Hard Carbon

Feasibility: Sustainable & Cost-Effective Anodes

1

Local Feedstock

- Cotton stalk, castor residue & groundnut shell.
- Low-cost, locally available carbon source.

2

Existing Infrastructure

- Use available heat/steam, power, N₂, gas & wastewater facilities.
- Lowers additional investment and setup cost.

3

Circular N/S Doping

- Recover N/S-containing industrial waste.
- Purify and use as doping precursor.
- Enables waste valorisation and circular economy.

4

Process & Performance

- Optimise carbonisation temperature, pore structure and interlayer spacing.
- Followed by N/S doping.
- Higher Na⁺ storage and improved rate performance.

5

Cost Advantage

- Compare imported vs locally produced hard carbon.
- Include biomass + processing + energy + doping + logistics costs.

6

Environmental Benefit

- Convert agro-waste & industrial waste into high-value battery material.
- Reduced waste and carbon footprint.

Sodium Precursor Sources in Gujarat

Abundant Resources | Local Supply Chain | Battery-Grade Potential

1

Common Salt (Sea Salt)

Gujarat's long coastline and salt pans provide abundant NaCl.

- Easily accessible and low cost
- Can be converted to Na₂CO₃ / NaOH (via brine route)
- Existing salt industry infrastructure

2

Soda Ash (Na₂CO₃)

Large production and availability in Gujarat (e.g., Ghogha, Kutch).

- Direct source of sodium
- Widely used in glass, detergents and chemicals
- Can be processed for battery-grade sodium salts

3

Brine (Underground / Saline Water)

Natural brine resources in Kutch and coastal regions.

- Rich in Na⁺ and other minerals
- Can be purified and converted to sodium carbonate / hydroxide
- Supports long-term, scalable supply

4

Industrial By-products

Sodium-containing waste streams from existing industries.

- Chlor-alkali (NaOH, Na₂CO₃)
- Glass and detergent industries (soda ash, silicates)
- Helps with circular economy and waste reduction

5

Other Potential Sources

Locally available mineral resources and agricultural residues.

- Sodium-rich minerals (e.g., trona, nahcolite – in some regions)
- Agro residues (e.g., plant ash, biomass) – for supplementary Na source
- Requires further exploration and process optimisation

Example Implementation

- 1. Feedstock Preparation:

Cotton stalk → cleaning & ash removal → drying → size reduction

- 2. Carbonisation:

Pre-carbonisation: 400–500°C → High-temperature

Carbonisation: 1,000–1,300°C, N₂/Ar → Hard Carbon

- 3. Purification & Yield :

De-ashing / washing → drying → particle-size control

Planning basis: 20–30% HC yield → 1 tonne dry biomass ≈ 200–300 kg HC

- 4. Performance Optimisation Baseline:

HC → XRD + Raman + BET + electrochemical testing → N/S doping using controlled N/S precursors → Post-treatment & purification → Final N/S-doped Hard Carbon

- 5. Battery Validation:

XRD + XPS + Raman → Half-cell testing → ICE + reversible capacity + cycling stability Literature Benchmark

Optimised cotton-derived HC: 345 mAh g⁻¹ reversible capacity | up to 95% ICE | 0.48 nm interlayer spacing

N-doped HC literature example: ~520 mAh g⁻¹ after 1000 cycles; ~204 mAh g⁻¹ at 1 A g⁻¹.



Hard Carbon Cost Comparison

IMPORTED vs. BIOMASS-DERIVED (Gujarat)

1 USD = ₹95.8918
(25 Sep 2026)

All costs converted to Indian Rupees using RBI reference rate (25 September 2026): ₹95.8918 / USD

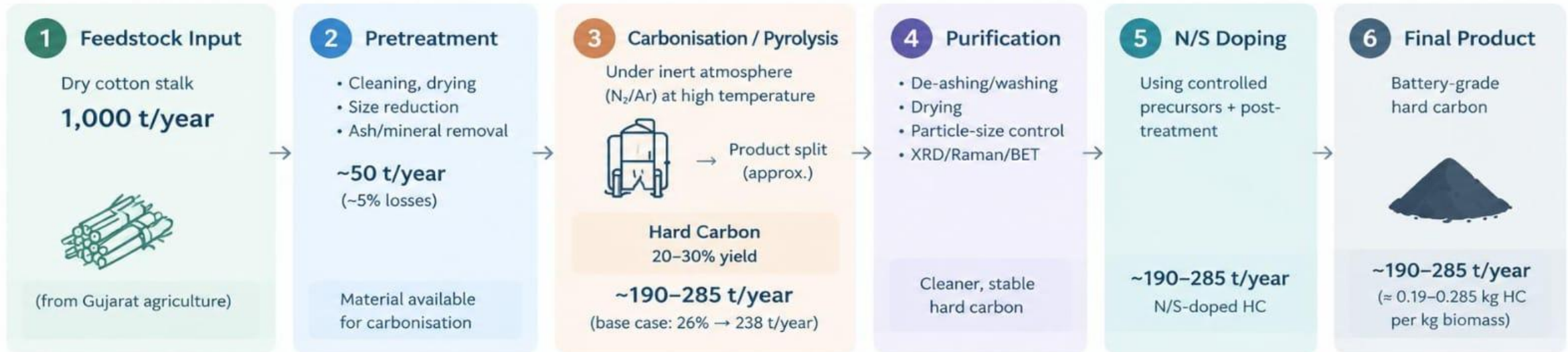
 Route		 Original Price (USD/kg)	 Price in INR/kg	 Price in INR/tonne	 Estimated HC material cost contribution per kWh (1.22 kg HC per kWh)
 IMPORTED HARD CARBON (Current market benchmark 2026)	Mainstream biomass-based HC from China	US\$8.40	₹805	₹8.05 lakh	₹982 / kWh
	Q3 2025 benchmark	US\$9.80	₹940	₹9.40 lakh	₹1,147 / kWh
	Premium coconut-shell HC	US\$26.20	₹2,512	₹25.12 lakh	₹3,065 / kWh
	Resin-based GWh-grade HC	US\$9.60	₹921	₹9.21 lakh	₹1,124 / kWh
 BIOMASS-DERIVED HARD CARBON (Published industrial TEA)	Biomass HC – normal wastewater treatment	US\$1.72	₹165	₹1.65 lakh	₹201 / kWh
	Biomass HC – wastewater recycling	US\$1.58	₹151.5	₹1.52 lakh	₹185 / kWh



Biomass to Hard Carbon – Mass Balance (Gujarat Pilot)

1,000 t/year dry cotton stalk (illustrative example)

*From agro-waste to
clean energy*



Existing infrastructure that can potentially be shared:

1. Heat/steam utilities → drying and thermal processing
2. Power infrastructure → grinders, pumps, furnaces, wastewater treatment
3. N₂/process-gas systems → inert atmosphere during carbonisation
4. Gas-treatment systems → treatment of pyrolysis/off-gases
5. Chemical storage/handling → pretreatment and doping chemicals



3-Month Roadmap

From Gujarat's Agricultural Residue to Sustainable Sodium-Ion Battery Anodes

Turning
waste into
energy 

1

Month 1

Feedstock & Process Validation
"Can we make it?"



Biomass Resource Assessment

- Identify local residues (cotton stalk, castor, groundnut shell)
- Check availability, seasonality, cost
- Initial transport & logistics estimate



Process Development

- Pretreatment (drying, size reduction)
- Optimise carbonisation conditions (T, time, atmosphere)



Infrastructure Study

- Map existing facilities (heat, power, N₂, wastewater)
- Feasibility of integration & cost estimate



Key Sources

Agricultural residue data (GOG, ICAR)
Process literature (e.g., biomass pyrolysis)
Local industry/plant databases

2

Month 2

Hard Carbon & Doping Validation
"Does it perform?"



Lab-scale Synthesis

- Carbonisation of selected biomass
- Optimise temperature, conditions
- HC yield & basic characterisation



Material Characterisation

- XRD, Raman, BET, SEM
- Structure, porosity, surface chemistry



N/S Doping

- Doping with N/S precursors (industrial waste)
- Optimise doping level & method



Electrochemical Testing

- Ionic conductivity, cycle stability
- Compare undoped vs N/S-doped HC



Key Sources

Scientific journals (e.g., Carbon, J. Mater. Chem.)
Characterisation labs (XRD, SEM, BET)
N/S precursor sources (industrial waste)

3

Month 3

Techno-Economic & Pilot Design
"Can we scale it economically?"



Cost & Economic Analysis

- Techno-economic comparison (HC vs graphite)
- Include raw material, processing, energy, doping, logistics
- Sensitivity analysis



District Selection & Logistics

- Choose district (based on biomass & infrastructure)
- Plan supply chain and transport route



Pilot Plant Design

- 1,000 t/year mass balance (preliminary)
- Equipment & area utility requirements
- Process flow & layout



Sustainability & Scale-up

- Environmental & carbon footprint
- Waste valorisation plan
- Scale-up feasibility



Key Sources

Gujarat industrial data (GIDC, DIC)
Cost databases (industry reports)
LCA tools / sustainability reports

References

Main Reference:

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Thank you!!